THE SOLUTIONS

1. (a) Evaluate the double integral

$$I \;=\; \int\int_R f(x,y)\; dy\; dx\;,$$

where $f(x,y) = yx^3 + xy^3$, and R is the region bounded by the lines $x = 0$ and $y = x$, and the curves $y^2 - x^2 = 2$ and $xy = 2$. To do so, begin by
(i) sketching the region R;
(ii) next obtain the values of the new variables $u = y^2 - x^2$ and $v = xy$ on the boundaries of R;
(iii) then show that the change of variables gives an integral with constant limits in the $uv-$plane;
(iv) finally, evaluate the integral. [9]

Solution: (i) Sketch at end. (ii) With new variables, the boundaries of R give us:

$x = 0 \Rightarrow v = xy = 0,$
$xy = 2 \Rightarrow v = 2,$
$y = x \Rightarrow u = y^2 - x^2 = 0,$
$y^2 - x^2 = 2 \Rightarrow u = 2.$

Hence with the change of variables, we have constant limits $u, v = 0 \ldots 2$, integrating over a square in the first quadrant of the $uv-$plane.

The Jacobian for the change of variables, $J_{u,v}(x,y) \;=\; \dfrac{\partial x}{\partial u}\dfrac{\partial y}{\partial v} - \dfrac{\partial y}{\partial u}\dfrac{\partial x}{\partial v}$ is difficult, but we can obtain

$$J_{u,v}(x,y) \;=\; \frac{1}{J_{x,y}(u,v)} \;=\; \frac{1}{\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} - \frac{\partial u}{\partial y}\frac{\partial v}{\partial x}} \;=\; \frac{1}{-2x(x) - 2y(y)} \;=\; -\frac{1}{2(x^2+y^2)}$$

With the Jacobian, we can write

$$f(x,y)\; dx\; dy \;=\; f(u,v)\; |J_{u,v}(x,y)|\; du\; dv \;=\; (yx^3 + xy^3)\frac{1}{2(x^2+y^2)}\; du\; dv \;=\; \frac{1}{2} v\; du\; dv$$

and integrate

$$I \;=\; \int_0^2 \int_0^2 \frac{1}{2} v\; dv\; du \;=\; \int_0^2 1\; du \;=\; 2\;.$$

(b) A periodic function $f(x)$ with period T, which satisfies the Dirichlet conditions, can be written as a complex Fourier Series:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nx/T} .$$

(i) Show that $\displaystyle\int_{-1}^{1} e^{ik\pi x}\, dx = \begin{cases} 0 & \text{for } k = \pm 1, \pm 2, \pm 3, \cdots \\ 2 & \text{for } k = 0. \end{cases}$. [3]

(ii) Hence, or otherwise, show that when $T = 2$, the complex Fourier coefficients are

$$c_n = \frac{1}{2}\int_{-1}^{1} f(x) e^{-in\pi x}\, dx$$

for every integer n. [5]

Solution:

(i)

$$\int_{-1}^{1} e^{ik\pi x}\, dx = \left[\frac{e^{ik\pi x}}{ik\pi}\right]_{-1}^{1} = \frac{1}{ik\pi}\left(e^{ik\pi} - e^{-ik\pi}\right) = \frac{2}{k\pi}\sin(k\pi) = 0\,,$$

for any non-zero integer k. For $k = 0$, we have $\displaystyle\int_{-1}^{1} e^0\, dx = 2$, trivially.

(ii) We take the function written as a series and multiply by $e^{-im\pi x}$:

$$f(x)e^{-im\pi x} = \left(\sum_{n=-\infty}^{\infty} c_n e^{in\pi x}\right) e^{-im\pi x} = \sum_{n=-\infty}^{\infty} c_n e^{i(n-m)\pi x} .$$

Now integrate both sides over one period:

$$\int_{-1}^{1} f(x)e^{-im\pi x}\, dx = \int_{-1}^{1} \left(\sum_{n=-\infty}^{\infty} c_n e^{i(n-m)\pi x}\right) dx = \sum_{n=-\infty}^{\infty} c_n \int_{-1}^{1} e^{i(n-m)x}\, dx$$

and as $n - m$ is an integer, every integral in the sum vanishes, except for the case $n = m \Rightarrow k = 0$, when the integral takes the value 2, as shown in (i), leaving only

$$\int_{-1}^{1} f(x)e^{-im\pi x}\, dx = 2c_m\,,$$

as required.

(c) A function is defined as $f(t) = t - 1$ for $0 \leq t < 1$. Obtain the complex Fourier series of the odd, period-2 extension of $f(t)$. [8]

Solution: Let the odd period-2 extension be $g(t)$. From (b), we have the formula for the complex coefficients:

$$c_n = \frac{1}{2}\int_{-1}^{1} f(t)e^{-in\pi t}\,dt\,,$$

but $g(t)$ takes different forms on the intervals $[-1,0]$ and $[0,1]$. We could do two integrals, but can reduce the amount of work by writing

$$c_n \;=\; \frac{1}{2}\int_{-1}^{1} g(t)[\cos(n\pi t) - i\sin(n\pi t)]\,dt \;=\; \frac{1}{2}\left(\int_{-1}^{1} g(t)\cos(n\pi t)\,dt \;-\; i\int_{-1}^{1} g(t)\sin(n\pi t)\,dt\right),$$

and as $g(t)$ is odd, the first integral is zero, and the second can be rewritten with the half-range approach, giving

$$c_n \;=\; i\int_{0}^{1} -(t-1)\sin(n\pi t)\,dt \;=\; i\left[(t-1)\frac{\cos(n\pi t)}{n\pi}\right]_0^1 - \frac{i}{n\pi}\int_0^1 \cos(n\pi t)\,dt\;.$$

The last integral gives $\frac{1}{n\pi}\sin(n\pi t)$, vanishing when $t = 0, 1$. This leaves $c_n \;=\; \dfrac{i}{n\pi}$. The series is therefore

$$g(t) \;=\; \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{i}{n\pi}\, e^{in\pi t}\,.$$

As $g(t)$ is odd, we don't need to calculate that $c_0 = 0$.

2. (a) Calculate the Fourier transform $F(\omega)$ of the function [7]

$$f(t) \;=\; \begin{cases} \cos(\pi t) & \text{for } |t| \le 1\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

Give careful consideration to values of ω where $F(\omega)$ has a zero denominator.

Solution: Write $f(t) = \dfrac{e^{i\pi t} + e^{-i\pi t}}{2}$ and proceed:

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-i\omega t}\, dt \;=\; \frac{1}{2}\int_{-1}^{1} \left(e^{i\pi t} + e^{-i\pi t}\right) e^{-i\omega t}\, dt \;=\; \frac{1}{2}\int_{-1}^{1} e^{-i(\omega-\pi)t} + e^{-i(\omega+\pi)t}\, dt$$

and integrate:

$$F(\omega) \;=\; \frac{1}{2}\left[\frac{e^{-i(\omega-\pi)t}}{-i(\omega-\pi)} + \frac{e^{-i(\omega+\pi)t}}{-i(\omega+\pi)}\right]_{-1}^{1} \;=\; \frac{e^{-i\omega} - e^{i\omega}}{2i(\omega-\pi)} + \frac{e^{-i\omega} - e^{i\omega}}{2i(\omega+\pi)}$$

given that $e^{\pm i\pi} = -1$, leaving

$$F(\omega) \;=\; -\sin(\omega)\left(\frac{1}{\omega-\pi} + \frac{1}{\omega+\pi}\right) \;=\; \frac{2\omega\sin(\omega)}{\pi^2-\omega^2}\,. \quad (*)$$

The denominator vanishes at $\omega = \pm\pi$, so we consider this separately:

$$F(\pi) \;=\; \frac{1}{2}\int_{-1}^{1} \left(e^{i\pi t} + e^{-i\pi t}\right) e^{-i\pi t}\, dt \;=\; \frac{1}{2}\left[t + \frac{e^{-2i\pi t}}{-2i\pi}\right]_{-1}^{1} \;=\; 1\,.$$

This is consistent with $(*)$ as

$$\lim_{\omega\to\pi} F(\omega) \;=\; \lim_{\omega\to\pi} \frac{\sin\omega}{\pi-\omega} \;=\; \lim_{z\to 0} \frac{\sin(\pi - z)}{z} \;=\; 1$$

substituting $z = \pi - \omega$, and using $\sin(\pi - x) = \sin x$, and similarly for $w = -\pi$.

(b) We are given a function of two variables $f(x,y) = (x+y)(x^2+y^2-1)$.

(i) Obtain the gradient and the Hessian determinant of $f(x,y)$ and use them to find and classify its stationary points. [6]

Solution: Write $f(x,y) = x^3+y^3+xy^2+x^2y-x-y,$ and differentiate to find grad$(f) =$

$(f_x, f_y) = (3x^2+y^2+2xy-1, 3y^2+2xy+x^2-1)$ and set both equal to zero, so that
$3x^2+y^2+2xy-1=0,$
$3y^2+2xy+x^2-1=0.$
Subtracting the two equations, simplifies to $2x^2-2y^2=0 \Rightarrow x=\pm y$. Substituting $x=y$ into the first equation (the second would work as well) we have

$$6x^2=1 \Rightarrow x=y=\pm\frac{1}{\sqrt{6}} \Rightarrow P_1=\left(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}}\right) \quad \text{and} \quad P_2=\left(-\frac{1}{\sqrt{6}},-\frac{1}{\sqrt{6}}\right)$$

are the first two stationary points. Substituting $x=-y$ into the first (or second) equation, we have

$$2x^2=1 \Rightarrow x=-y=\pm\frac{1}{\sqrt{2}} \Rightarrow P_3=\left(\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right) \quad \text{and} \quad P_4=\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$$

Now find the second derivatives: $f_{xx}=2(3x+y)$, $f_{yy}=2(x+3y)$, and $f_{xy}=2(x+y)$ so the Hessian determinant is

$$D=f_{xx}f_{yy}-(f_x)^2=4(3x+y)(x+3y)-4(x+y)^2=8(x^2+y^2+4xy)$$

Evaluating at $P_{1,2}$, we have $D=8(\frac{1}{6}+\frac{1}{6}+\frac{4}{6})>0$, so extrema. For $P_1, f_{xx}=8/\sqrt{6}>0$ so a minimum , and for $P_2, f_{xx}=-8/\sqrt{6}<0$, so a maximum.

Evaluating at $P_{3,4}$ we have $D=8(\frac{1}{2}+\frac{1}{2}-\frac{4}{2})<0$, so these are saddle points.

(ii) Draw the zero contour and locate the stationary points. Sketch the contour plot, explaining your reasoning. [5]

Solution: For the zero contour, we set $f(x,y)=(x+y)(x^2+y^2-1)=0 \Rightarrow y=-x$, or $x^2+y^2=1$. These divide the plane into four regions, labelled $I-IV$ in the sketch. As P_1 is a minimum, the function takes negative values in region I; likewise, as P_2 is a maximum, the function takes positive values in region II. Moving from P_1 in a north-east direction, crossing from I to III, the values increase from negative to zero to positive, by continuity, so the values in III are positive and the values of the function go to $+\infty$. Moving from P_2 in a south-west direction, crossing from Ii to IV, the values increase from positive to zero to negative, by continuity, so the values in IV are negative and the values of the function go to $-\infty$. Sketch at end.

(c) Find the function $g(y)$ which makes the following first-order ordinary differential equation exact:

$$(x\sin y + g(y) + xe^{xy})\frac{dy}{dx} + ye^{xy} + 2x + g(y) = 0\,.$$

Hence obtain the solution of the differential equation satisfying $y = 0$ when $x = 1$. [7]

Solution: Identify $P(x,y) = x\sin y + g(y) + xe^{xy}$ and $Q(x,y) = ye^{xy} + 2x + g(y)$. To be exact, we require $\dfrac{\partial P}{\partial x} = \dfrac{\partial Q}{\partial y}$. So

$$\sin y + (1+xy)e^{xy} \;=\; \frac{\partial P}{\partial x} \;=\; \frac{\partial Q}{\partial y} \;=\; (1+xy)e^{xy} + g'(y)$$

so to be exact, we need $g'(y) = \sin y \Rightarrow g(y) = -\cos y$. We can now find $f(x,y)$ so that the ODE is equivalent to $df = 0$ with

$$\frac{\partial f}{\partial x} \;=\; Q \Rightarrow f \;=\; \int Q\,dx \;=\; \int ye^{xy} + 2x - \cos y\,dx \;=\; e^{xy} + x^2 - x\cos y + A(y)$$

where $A(y)$ is an arbitrary function, and

$$\frac{\partial f}{\partial y} \;=\; P \Rightarrow f \;=\; \int P\,dy \;=\; \int x\sin y - \cos y + xe^{xy}\,dy \;=\; -x\cos y - \sin y + e^{xy} + B(x)$$

where $B(x)$ is an arbitrary function. Equating the two expressions, we have $A(y) = -\sin y$ and $B(x) = x^2$. The solution is $f = constant$ giving

$$-x\cos y - \sin y + e^{xy} + x^2 = C$$

The condition $x = 1 \Rightarrow y = 0$ gives $C = 1$.

3. (a) Show that the Laplace transform of the third derivative of a function is as follows: [5]

$$\mathscr{L}[f'''(t)] = s^3\overline{f}(s) - s^2 f(0) - sf'(0) - f''(0)\,.$$

Solution:

Let $g(t) = f''(t)$, the second derivative. Then $f''' = g'$ and $\mathscr{L}[f'''(t)] \;=\; \mathscr{L}[g'(t)] \;=\; s\overline{g}(s) - g(0)(*)$

Given that $g(t) = f''(t)$, we immediately have $g(0) = f''(0)$, and the transform of $g(t)$ is

$$\overline{g}(s) \;=\; \mathscr{L}[g(t)] \;=\; \mathscr{L}[f''(t)] \;=\; s^2\overline{f}(s) - sf(0) - f'(0)$$

as in the formula sheet. Substituting into $(*)$ we get

$$\mathscr{L}[f'''(t)] \;=\; s\left[s^2\overline{f}(s) - sf(0) - f'(0)\right] - f''(0) \;=\; s^3\overline{f}(s) - s^2 f(0) - sf'(0) - f''(0)\;,$$

as required.

(b) Use (a) and Laplace transforms to show that the solution of the third-order ODE

$$\frac{d^3x}{dt^3} + \frac{dx}{dt} = \begin{cases} 1 & \text{for} \quad 0 \le t < \pi \\ 0 & \text{for} \quad \pi \le t \end{cases}$$

which satisfies the initial conditions $x''(0) = -1\;, x'(0) = 1\;, x(0) = 1$, is given by

$x(t) = \begin{cases} x_1(t) = t + \cos t & \text{for} \quad 0 \le t < \pi\;, \\ x_2(t) & \text{for} \quad \pi \le t\;, \end{cases}$ giving the correct form of $x_2(t)$. [10]

Solution: The r.h.s of the ODE is written in terms of the Heaviside function

$$\begin{cases} 1 & \text{for} \quad 0 \le t < \pi \\ 0 & \text{for} \quad \pi \le t \end{cases} \;=\; 1 - H(t-\pi)$$

and now take Laplace transforms of both sides:

$$s^3\bar{x}(s) - s^2x(0) - sx'(0) - x''(0) + s\bar{x}(s) - x(0) = \frac{1}{s}(1 - e^{-\pi s})$$

include inititial conditions and rearrange: $s(s^2+1)\bar{x} = s^2 + s + \frac{1}{s}(1 - e^{-\pi s})$

and divide by $s(s^2+1)$ to get $\bar{x} \;=\; \dfrac{s+1}{s^2+1} + \dfrac{1}{s^2(s^2+1)}(1 - e^{-\pi s}) \;=\; \dfrac{s+1}{s^2+1} + \left(\dfrac{1}{s^2} - \dfrac{1}{s^2+1}\right)(1 - e^{-\pi s})$

using partial fractions, which simplifies to

$$\bar{x} \;=\; \frac{s}{s^2+1} + \frac{1}{s^2} - \left(\frac{1}{s^2} - \frac{1}{s^2+1}\right)e^{-\pi s}$$

which is ready to take the inverse Laplace transform:

$$x(t) = \cos t + t - [t - \sin t]_{t \to t-\pi} H(t-\pi)$$

using the second-shift theorem, giving

$$x(t) \;=\; \begin{cases} t + \cos t & \text{for} \quad 0 \le t < \pi\;, \\ t + \cos t - [t - \pi - \sin(t-\pi)] & \text{for} \quad \pi\;, \le t \end{cases} \;=\; \begin{cases} t + \cos t & \text{for} \quad 0 \le t < \pi\;, \\ \pi + \cos t - \sin t & \text{for} \quad \pi \le t\;, \end{cases}$$

as required.

(c) Given the matrices

$$A = \begin{pmatrix} 3 & -3 & 2 \\ -1 & 3 & -1 \\ -2 & \alpha & -2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 4 & 1 & 3 \\ 3 & -2 & 4 \\ -2 & 3 & -2 \end{pmatrix}$$

(i) Find the value of α for which A^{-1} does not exist, [3]

Solution: Expanding by the third row, the determinant is

$$|A| = -2(3-6) - \alpha(-3+2) - 2(9-3) = \alpha - 6$$

so the matrix will not be invertible if $\alpha = 6$.

(ii) If $\alpha = 4$ find $\det(A)$, $\det(B)$, and $\det(AB)$, [3]

Solution: Given (i), $\det(A) = 4-6 = -2$. Obtain

$$|B| = 4(4-12) - 1(-6+8) + 3(9-4) = -19\ .$$

Hence $\det(AB) = \det(A)\det(B) = -2(-19) = 38$.

(iii) If $\alpha = 7$, use Gauss-Jordan elimination to calculate A^{-1}. [4]

Solution: Use Gauss-Jordan elimination to get $(A : I) \sim (I : A^{-1})$:

$$\left(\begin{array}{ccc|ccc} 3 & -3 & 2 & 1 & 0 & 0 \\ -1 & 3 & -1 & 0 & 1 & 0 \\ -2 & 7 & -2 & 0 & 0 & 1 \end{array}\right) \underset{-R_2 \leftrightarrow R_1}{\sim} \left(\begin{array}{ccc|ccc} 1 & -3 & 1 & 0 & -1 & 0 \\ 3 & -3 & 2 & 1 & 0 & 0 \\ -2 & 7 & -2 & 0 & 0 & 1 \end{array}\right) \underset{\substack{R_2-3R_1 \\ R_3+2R_1}}{\sim} \left(\begin{array}{ccc|ccc} 1 & -3 & 1 & 0 & -1 & 0 \\ 0 & 6 & -1 & 1 & 3 & 0 \\ 0 & 1 & 0 & 0 & -2 & 1 \end{array}\right)$$

$$\underset{R_2 \leftrightarrow R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & -3 & 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & -2 & 1 \\ 0 & 6 & -1 & 1 & 3 & 0 \end{array}\right) \underset{\substack{R_1+3R_2 \\ R_3-6R_2}}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & -7 & 3 \\ 0 & 1 & 0 & 0 & -2 & 1 \\ 0 & 0 & -1 & 1 & 15 & -6 \end{array}\right) \underset{-R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & -7 & 3 \\ 0 & 1 & 0 & 0 & -2 & 1 \\ 0 & 0 & 1 & -1 & -15 & 6 \end{array}\right)$$

$$\underset{R_1 - R_3}{\sim} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 8 & -3 \\ 0 & 1 & 0 & 0 & -2 & 1 \\ 0 & 0 & 1 & -1 & -15 & 6 \end{array}\right) \text{ so we have } A^{-1} = \begin{pmatrix} 1 & 8 & -3 \\ 0 & -2 & 1 \\ -1 & -15 & 6 \end{pmatrix}.$$

4. (a) You are given the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 2 & 5 & 7 \end{pmatrix}$.

(i) Use Gaussian elimination to show why we cannot write $A = LU$, where L is lower-triangular and U upper-triangular. Explain your reasoning. [5]

Solution: To obtain an upper triangular matrix U we use row-operations:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 2 & 5 & 7 \end{pmatrix} \overset{\sim}{R_2 - R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 2 & 5 & 7 \end{pmatrix} \overset{\sim}{R_3 - 2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \overset{\sim}{R_3 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = U$$

The last matrix is upper-triangular, but because a row-swap was involved, we cannot write $A = LU$. We can represent the row operations in terms of two elimination matrices and one permutation matrix, whose product is E, where $EA = U$. Because E included a permutation matrix, it is not lower-triangular, and its inverse will not be lower triangular, either. (Alternative explanation possible)

(ii) Your answer to (i) should have included a single row exchange. Write down the permutation matrix P, with the property that PA can be written as the product of a lower-triangular matrix L, and an upper-triangular matrix U. Now use Gaussian elimination to find U.

With no further calculation, write down the two elimination matrices involved, and hence write down L. [5]

Solution: The permuation matrix is $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$, representing $R_2 \leftrightarrow R_3$. Hence

$$PA = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 2 & 4 \end{pmatrix} \overset{\sim}{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix} \overset{\sim}{R_3 - R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} = U$$

The elimination matrices are $E_{21} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $E_{31} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$,

which immediately gives $L = (E_{31}E_{21})^{-1} = E_{21}^{-1}E_{31}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

(b) Given a 4×3 matrix A, what can we say about rank, nullspace, column space and solutions to $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$, without any calculations? [5]

Solution: The rank is at most $r = 3$. With full column rank, $r = 3$, every column contains a pivot and there will be no free variables, so the nullspace will consist of the zero vector. In this case the columns are linearly independent and form a basis for the column space, a subspace of $\mathbb{R}^4$.

If the rank is less than three, the column space will be a plane $(r = 2)$ or a line $(r = 1)$ in $\mathbb{R}^4$. As the rank decreases, the nullspace increases with the number of free variables: for $r = 2$, the nullspace will be a line, for $r = 1$, a plane, both through the origin, subspaces of $\mathbb{R}^3$.

Solutions to $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ exist only if $\underline{\mathbf{b}}$ is in the column space. If they exist, when $r = 3$ the solution will be unique, and if $r < 3$ there will be infinitely many solutions: when $r = 2$, a line or, when $r = 1$, a plane, both times parallel to the nullspace, but not through the origin.

Full marks, even without considering the extreme case $r = 0$, when the column space will consist of the zero vector alone, the nullspace is $\mathbb{R}^3$ and solutions exist only if $\underline{\mathbf{b}}$ is the zero vector.

(c) Given

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix},$$

obtain the reduced echelon form of A. State rank, and free and pivot columns and variables. Hence find bases for the four fundamental subspaces of A: nullspace, column space, row space, and left nullspace. In each case, state the dimension and the parent space, as well as a geometric description, if possible. [10]

Solution: Use Gaussian elimination to find the reduced echelon form:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} \underset{\substack{R_3-7R_1 \\ R_4-10R_1}}{\sim} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \\ 0 & -9 & -18 \end{pmatrix} \underset{-\frac{1}{3}R_2}{\sim} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -6 & -12 \\ 0 & -9 & -18 \end{pmatrix} \underset{\substack{R_3+6R_2 \\ R_4+9R_2}}{\sim} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = R$$

is the reduced echelon form. We note that the first two columns are pivot columns, and the third column is free, so the rank is two. Likewise, x_1 and x_2 are pivot variables, while x_3 is a free variable. A basis for the column space is therefore given by the first two columns of A, which are $(1,4,7,10)^T$ and $(2,5,8,11)^T$. The two-dimensional column space is therefore a plane in $\mathbb{R}^4$.

The two non-zero rows of R are $(1,0,-1)^T$ and $(0,1,2)^T$ and give a basis for the two-dimensional row space, a plane in $\mathbb{R}^3$.

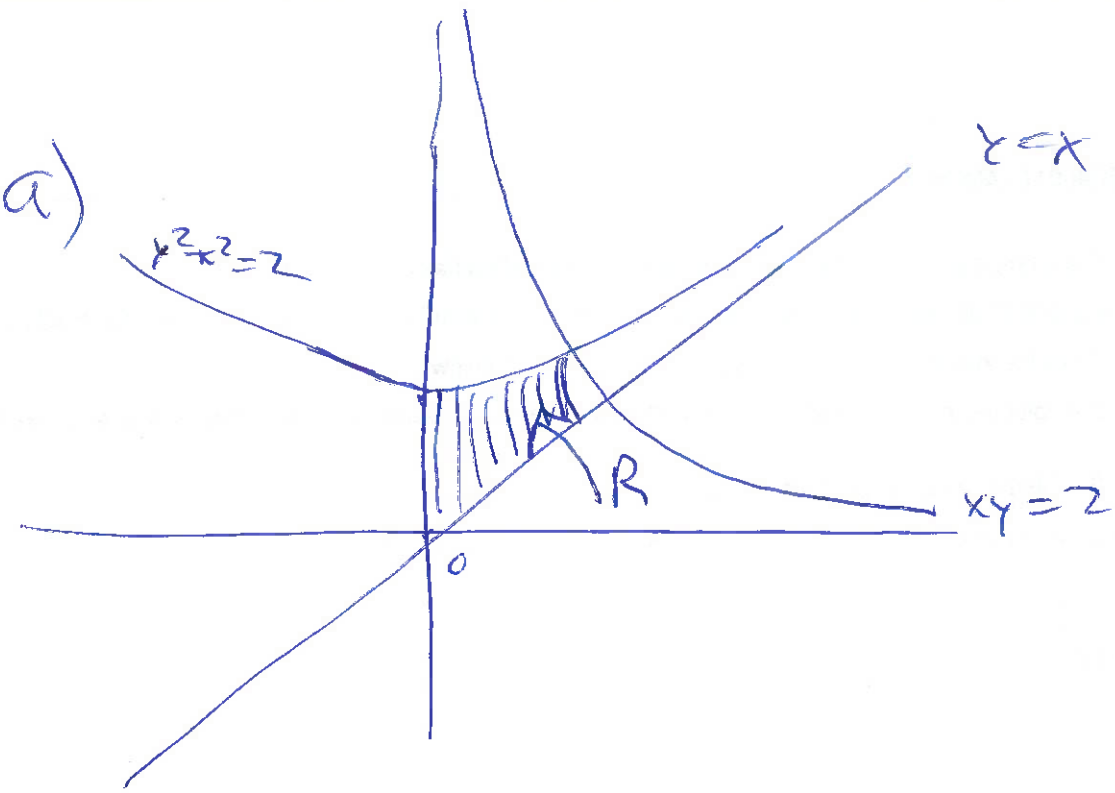
There is one free variable, so let $x_3 = 1$. The nullspace is solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{0}}$. The top two rows give equations $x_1 - x_3 = 0$ and $x_2 + 2x_3 = 0$, so with $x_3 = 1$ we immediately have $x_1 = 1$ and $x_2 = -2$ giving the special solution $\underline{\mathbf{x}}^s = (1,-2,1)^T$, a basis for the one-dimensional nullspace, a line in $\mathbb{R}^3$.

To find a basis for the left nullspace, we find a matrix E by applying to I_4 the row operations which gave R. As there are two zero rows in R, the left nullspace will be two-dimensional, and the last two rows of E will be the basis. Hence I_4

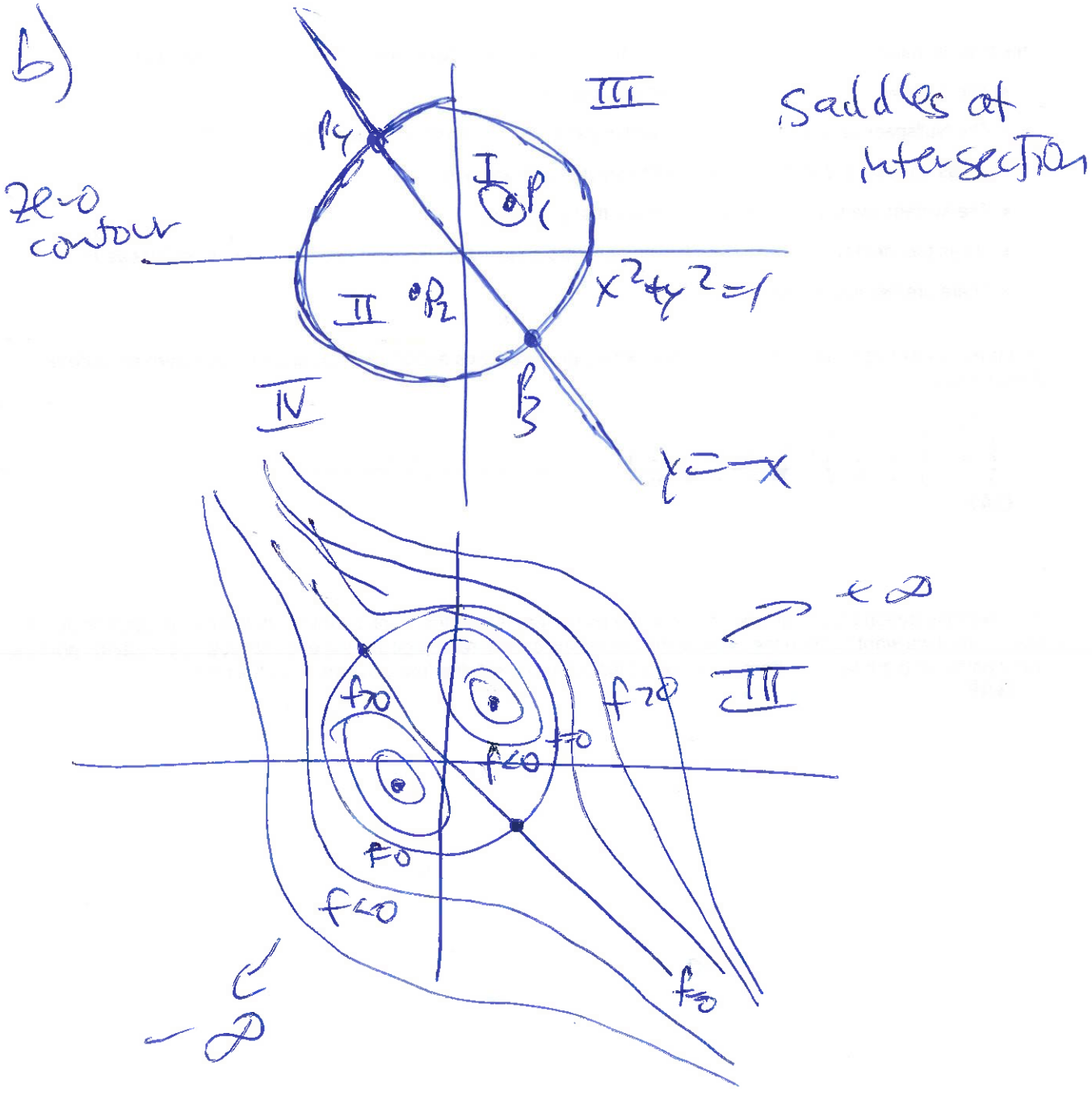
$$\underset{\substack{R_2-4R_1 \\ R_3-7R_1 \\ R_4-10R_1}}{\sim} \begin{pmatrix} 1 & 0 & 0 & 0 \\ -4 & 1 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ -10 & 0 & 0 & 1 \end{pmatrix} \underset{-\frac{1}{3}R_2}{\sim} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 4/3 & -1/3 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ -10 & 0 & 0 & 1 \end{pmatrix} \underset{\substack{R_3+6R_2 \\ R_4+9R_2}}{\sim} \begin{pmatrix} -5/3 & 2/3 & 0 & 0 \\ 4/3 & -1/3 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 2 & -3 & 0 & 1 \end{pmatrix} = E$$

The matrix E is constructed from the row operations by which $A \sim R$, so as matrix multiplication, $EA = R$. The last two rows of E represent linear combinations of the rows of A which give the zero rows in R:
$1(1,2,3) - 2(4,5,6) + 1(7,8,9) + 0(10,11,12) = (0,0,0)$,
$2(1,2,3) - 3(4,5,6) + 0(7,8,9) + 1(10,11,12) = (0,0,0)$.
Any vector in the left nullspace satisfies $A^T\underline{\mathbf{y}} = \underline{\mathbf{0}} \Rightarrow \underline{\mathbf{y}}^T A = \underline{\mathbf{0}}^T$, so the last two rows of E are such vectors, and the basis for the two-dimensional left nullspace is given by $(1,-2,1,0)^T$ and $(2,-3,0,1)^T$, giving a plane in $\mathbb{R}^4$.

1) a)


y=x
x²+y²=2
R
xy=2
0

2) b)


III
Saddles at intersection
zero contour
I
II
IV
x²+y²=1
y=-x
f>0
f<0
f=0
-∞
+∞